

Full Carlitz Torsion Dynamics and Ramified Prime-Power Towers

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Abstract

We study the full Carlitz ring action over $\mathbb{F}_q(\theta)$, for every prime power q , rather than one additive polynomial in fixed characteristic. Every conductor torsion module is cyclic, and its annihilator strata give the exact preperiod and eventual period of every multiplier. Fixed-point counts and finite determinants follow without treating nonunit multipliers as permutations. An annihilator argument proves the specific reduction $C_P(X) \bmod P = X^{q^{\deg P}}$. Primitive polynomials at every prime-power level are Eisenstein. Their roots generate the full unit-group Galois action; finite-prime ramification proves linear disjointness and the corresponding theorem at every conductor. Compatible towers realize completed local unit groups. Explicit integral bases give all lower ramification groups and two matching formulas for the different. Independent finite-ring orbit enumeration audits 107 conductors and 77 prime-power cases, including a nonprime constant field. The source theorem is a synthesis of classical Carlitz cyclotomic theory, not a literature-priority claim or a target Riemann spectral construction.

Keywords: Carlitz module; torsion dynamics; annihilator strata; prime-power towers; ramification; source determinant.

中文摘要

本文研究一般特征下的完整 Carlitz 环作用，以全部导子扭点模为相空间，给出每个乘子作用的精确前周期、最终周期与原生有限行列式。关键约化由消去理想与不同根数矛盾证明，不把任意加性多项式误当 Frobenius。所有素多项式幂层的原始多项式均满足 Eisenstein 判据，进而得到完整 Galois 单位群、不同素导子塔的线性无交与任意导子结论。进一步计算兼容无限塔、下编号分歧群及 different 指数，并以独立有限商环枚举核验。这些结果属于经典源系统理论的自包含整合，不宣称文献首创，也不提供目标零点或量子构造。

关键词：Carlitz 模；扭点动力学；消去理想分层；素幂塔；分歧；原生行列式。

1 Frozen source and the complete torsion dynamics

Put $A = \mathbb{F}_q[\theta]$, $K = \mathbb{F}_q(\theta)$, and $C_\theta(X) = \theta X + X^q$. The base field is exactly K . For $a = \sum a_j \theta^j$, define $C_a = \sum a_j C_\theta^{e_j}$. Constants from $\mathbb{F}_q$ commute with C_θ , so $C_{a+b} = C_a + C_b$ and $C_{ab} = C_a \circ C_b = C_b \circ C_a$. For monic nonzero a , induction gives $\deg_X C_a = q^{\deg a}$ and $C'_a = a \neq 0$. Thus $\Lambda_a = \ker C_a \subset K^{\text{sep}}$ has exactly $q^{\deg a}$ distinct points. Generic characteristic refers to the injective coefficient map $A \rightarrow K$; the field still has characteristic p when $q = p^e$. Reduced torsion means multiplicity one, not Lyapunov stability.

The classical ring and cyclotomic framework is discussed in Conrad's author exposition [1] and in the tower background of Anglès–Bandini–Bars–Longhi [2]. The following proofs do not use their Iwasawa main conjecture. Our contribution within this repository is an all-conductor ring-action and ramification package: C384 studied a single map $x + x^p$ and its wild geometric zeta, while general finite-map determinants were already available in C204. Those finite formulas alone are not credited as the new mechanism.

Round 0 advance. We close the full annihilator/clock classification and the critical prime-reduction lemma, including nonunit and zero multipliers.

Theorem 1 (All conductor modules and multiplier clocks). *For every monic nonzero a , $\Lambda_a \cong A/a$ as an A -module. Points of exact annihilator $D \mid a$ have population $\Phi(D) = |(A/D)^\times|$, with $\Phi(1) = 1$. For $b \neq 0$, define*

$$D_* = \prod_{P \mid b} P^{v_P(D)}, \quad h_D = \max_{P \mid b} \left\lceil \frac{v_P(D)}{v_P(b)} \right\rceil.$$

The empty maximum is zero. Under C_b , a point in this stratum has exact preperiod h_D and eventual period $\text{ord}_{D_}(b)$, with $\text{ord}_1(b) = 1$. For $b = 0$, every nonzero point reaches the fixed point zero in one step.*

Proof. For P irreducible and $k \geq 1$, put $Q = q^{\deg P}$. There are Q^k points in Λ_{P^k} but only Q^{k-1} in $\Lambda_{P^{k-1}}$. Choose λ_k in their difference. Its annihilator ideal contains (P^k) but not P^{k-1} , hence is (P^k) . The injection $A/P^k \rightarrow \Lambda_{P^k}$ sending r to $C_r(\lambda_k)$ is bijective by cardinality. If u, v are coprime and $ru + sv = 1$, then $x = C_{sv}(x) + C_{ru}(x)$ splits Λ_{uv} into its u and v torsion, with zero intersection by Bezout. Induction proves the CRT and cyclicity. Exact-annihilator- D points are the classes $(a/D)r$ with $r \in (A/D)^\times$, giving their stated count.

The annihilator after one step is $D/\gcd(D, b)$, because $C_c(C_b x) = 0$ if and only if $D \mid cb$. After j steps the exponent at $P \mid b$ is $\max(v_P(D) - jv_P(b), 0)$, whereas other exponents are unchanged. A strictly decreasing annihilator cannot occur on a cycle. Exactly at h_D the remaining annihilator is D_* , where b is a unit. Return after n steps is then equivalent to $D_* \mid b^n - 1$. The assertions for $b = 0$, for $b = 1$, and for the one-point module $a = 1$ follow directly from $C_0 = 0$ and $C_1 = X$. $\square$

Proposition 1 (Finite source counts and determinant). *For $n \geq 1$ and every $b \in A$,*

$$F_n(a, b) = \# \text{Fix}(C_b^n \mid \Lambda_a) = q^{\deg \gcd(a, b^n - 1)}, \quad O_n = \frac{1}{n} \sum_{d \mid n} \mu(n/d) F_d.$$

For pullback U_b on all complex functions on Λ_a ,

$$\det(I - zU_b) = \prod_{d \geq 1} (1 - z^d)^{O_d}, \quad \exp \left(\sum_{n \geq 1} F_n z^n / n \right) = \det(I - zU_b)^{-1}.$$

Proof. The kernel of multiplication by $c = b^n - 1$ on A/a consists of the multiples of $a/\gcd(a, c)$, giving its size. Set $\gcd(a, 0) = a$. Every finite functional graph is a disjoint union of directed cycles with incoming trees, so $F_n = \sum_{d \mid n} dO_d$ and Möbius inversion gives integer nonnegative O_d . Ordering vertices by distance to a cycle makes the point-map matrix block triangular: the transient block is nilpotent, and a length- d cycle contributes $1 - z^d$. Transposing gives the pullback formula; expansion of $-\log(1 - z^d)$ proves the formal zeta identity. $\square$

In particular, a nonunit multiplier is not generally a permutation of the whole torsion module. The two clocks here are multiplier iteration n and conductor depth k ; neither is declared to be a rational-prime logarithm.

2 The critical Carlitz reduction

Lemma 1. *For every monic irreducible P , reduction of coefficients modulo P gives $\overline{C}_P(X) = X^Q$, where $Q = q^{\deg P}$.*

Proof. The reduced polynomial is monic of degree Q and has derivative zero. Suppose it has a nonzero root γ over $\overline{\mathbb{F}}_Q$. The map $a \mapsto \overline{C}_a(\gamma)$ has a proper annihilator ideal: it contains (P) , while $C_1(\gamma) = \gamma \neq 0$. Maximality of (P) identifies this annihilator as exactly (P) , so the ring orbit has Q distinct elements. Commutation makes them all roots of $\overline{C}_P$. But a degree- Q polynomial with derivative zero in characteristic p has every root of multiplicity at least p , and cannot have Q distinct roots. Thus zero is the only root; monicity gives X^Q . $\square$

This proves a specific ring-action property. It is not an assumption that an arbitrary additive polynomial reduces to Frobenius.

3 All prime-power Galois towers and composite conductors

Round 1 advance. We add primitive irreducibility, full Galois surjectivity, valuation ownership, and a finite-prime proof of disjointness at arbitrary conductors.

Theorem 2. For $k \geq 1$, let $e_k = (Q - 1)Q^{k-1}$ and

$$\Psi_P = C_P/X, \quad \Psi_{P^k} = \Psi_P \circ C_{P^{k-1}}.$$

This is a monic P -Eisenstein polynomial of degree e_k whose roots are exactly the primitive P^k -torsion points. For any such root λ_k ,

$$L_{P^k} = K(\lambda_k), \quad \text{Gal}(L_{P^k}/K) = (A/P^k)^\times, \quad \sigma_u(\lambda_k) = C_u(\lambda_k).$$

The prime P is totally ramified, $v_P(\lambda_k) = 1/e_k$, and every other finite prime is unramified. For all monic a , $\text{Gal}(L_a/K) = (A/a)^\times$.

Proof. The identity $C_P(Y) = Y\Psi_P(Y)$ gives $C_{P^k} = C_{P^{k-1}}\Psi_{P^k}$. All roots of the left side are simple, so the quotient has precisely the primitive roots. Reduction gives $\overline{\Psi}_{P^k} = X^{e_k}$, while its constant coefficient is exactly P . Thus every nonleading coefficient is divisible by P , but the constant coefficient is not divisible by P^2 . Eisenstein proves irreducibility. A primitive root generates the entire torsion module, so its field is the splitting field. Galois acts faithfully by units; the irreducible degree equals the number of units, proving surjectivity.

At a place over P , an Eisenstein root has positive valuation. Otherwise the monic term is uniquely minimal. With positive valuation every nonconstant, nonleading term has valuation greater than one; cancellation forces $e_k v_P(\lambda_k) = 1$. Its denominator forces ramification index at least e_k , and the global degree bounds it above. Hence the prime has one place, residue degree one, and total ramification.

At a finite $R \neq P$, the monic polynomial C_{P^k} has derivative P^k , a unit. Its residue roots are distinct in a finite residue extension. Pass to the associated unramified extension of the complete field K_R . Hensel lifting gives all the roots there, so the splitting field is unramified at R .

Now form a compositum of distinct-prime towers inductively. At the next prime P , the previous compositum is unramified, while the new tower is totally ramified. Their intersection is both. A subextension of a globally totally ramified extension has one place with local degree equal to global degree; an extension both total and unramified has degree one. Hence the intersection is K . The two Galois extensions are linearly disjoint. Multiplying degrees and using the module CRT identifies the compositum with L_a and gives exactly $(A/a)^\times$. $\square$

Infinity ramification and the full maximal abelian extension are outside this theorem; constant extensions are not silently added.

4 Compatible towers, lower groups, and the different

Round 2 advance. We complete the infinite tower and local integer-ring argument, derive all ramification jumps, and confront source-to-target promotion with controls.

Choose compatible primitive roots by solving $C_P(\lambda_{k+1}) = \lambda_k$. Such a root is killed by P^{k+1} but not by P^k , since otherwise $C_{P^{k-1}}(\lambda_k) = 0$. Restriction of automorphisms is unit reduction modulo P^k , which is surjective. Compatible restrictions glue on the union, so as profinite groups

$$\text{Gal}\left(\bigcup_{k \geq 1} L_{P^k}/K\right) = \varprojlim_k (A/P^k)^\times = A_P^\times.$$

The infinite quantifier follows from these compatible algebraic maps, not from a finite truncation.

Theorem 3 (Exact lower filtration and different). *Normalize $v_{L_k}(\lambda_k) = 1$. For $u \neq 1$ and $s = v_P(u - 1) < k$, the ramification displacement is $v_{L_k}(\sigma_u \lambda_k - \lambda_k) = Q^s$. Consequently*

$$\begin{aligned} G_0 &= (A/P^k)^\times, \\ G_i &= 1 + P^s \pmod{P^k} \quad (Q^{s-1} \leq i \leq Q^s - 1, \ 1 \leq s < k), \\ G_i &= \{1\} \quad (i \geq Q^{k-1}), \quad d_k = Q^{k-1}(k(Q-1) - 1). \end{aligned}$$

Proof. Write $u - 1 = P^s v$ with v a unit. Then $\sigma_u \lambda_k - \lambda_k = C_v(C_{P^s} \lambda_k)$ is primitive of level $k - s$, so its base valuation is $1/e_{k-s}$ and its L_k valuation is $e_k/e_{k-s} = Q^s$.

We must justify testing ramification on one generator. The powers $1, \lambda_k, \dots, \lambda_k^{e_k-1}$ form a field basis. In an expansion $x = \sum_j a_j \lambda_k^j$, nonzero summands have valuations $e_k v_P(a_j) + j$, distinct modulo e_k . The minimum is unique. Thus x is integral exactly when all a_j are integral, proving $\mathcal{O}_{L_k} = \mathcal{O}_{K_P}[\lambda_k]$. The difference of any integral

polynomial in λ_k under σ is divisible by $\sigma\lambda_k - \lambda_k$ with integral quotient. The minimum difference valuation is attained on λ_k itself. The lower-group definition $v_{L_k}(\sigma x - x) \geq i + 1$ for every integral x now gives the stated intervals and group sizes.

Summing their contributions gives

$$d_k = (e_k - 1) + \sum_{s=1}^{k-1} (Q^s - Q^{s-1})(Q^{k-s} - 1) = Q^{k-1}(k(Q - 1) - 1).$$

Independently, differentiate $C_{P^k} = C_{P^{k-1}}\Psi_{P^k}$ at λ_k to get $P^k = \lambda_1\Psi'_{P^k}(\lambda_k)$. The monogenic different formula therefore yields $ke_k - Q^{k-1}$, the same answer. When $k = 1$ the intermediate intervals are empty; when also $Q = 2$, the degree is one, the group is trivial and the different exponent is zero. $\square$

5 Evidence, controls, and HEN-O373

The exact grid covers $q = 2, 3, 4, 5$. Conductors have degree at most three for $q = 2, 3$, and at most two for $q = 4, 5$: 107 cases in total. Every residue multiplier is included, with its complete preperiod/period population and fixed counts through time twelve. The 77 prime-power cases use irreducibles of degree at most three for $q = 2$ and at most two otherwise, $1 \leq k \leq 3$, and $q^{k \deg P} \leq 256$. The nonprime field is $\mathbb{F}_4 = \mathbb{F}_2[w]/(w^2 + w + 1)$.

The producer uses annihilator formulas and left composition. The independent checker imports no producer code: it enumerates every finite functional graph, constructs annihilators from residue classes, uses right composition, and obtains primitive polynomials by exact division. It also enumerates unit displacements for the ramification histogram. SymPy checks exact polynomial, finite-matrix, field-table and different identities. Two fresh working directories replay the same frozen producer source. Repaired-hash mutations test semantic and exact-type failures; these are not statistical tests or independent copies of the source tree.

Three structural controls locate the arithmetic carrier. The rank-zero parent $C_a(X) = aX$ has no nonzero torsion. A distinct-prime conductor PQ has a CRT product, while P^2 has a principal-unit filtration. Finally, $\theta X + uX^q$ is geometrically conjugate to the frozen rank-one action: choose v with $uv^{q-1} = 1$ and conjugate by $X \mapsto vX$. This conjugacy need not descend to K , so arbitrary twists do not inherit the base-field Galois theorem without further work.

The strict Route A tuple is structural arithmetic relation at A0, weak A1, and failures at A2, A3 and A4; the overall judgement is exploratory. The source owns polynomial primes, exact torsion and ramification. It does not identify them with rational primes or construct logarithmic rational-prime orbit clocks. None of the six mandatory strong-A1 controls is complete. Source finite determinants have no target divisor ownership; no target completion, functional equation, counting law, Weil compression, quantization or Hilbert–Polya domain is supplied. Route B remains disabled.

6 Conclusion

The full ring action, rather than an isolated additive-map census, makes annihilator ideals and coefficient primes intrinsic to the source. Primitive Eisenstein polynomials close the full finite Galois and composite-conductor construction. Compatibility and integral local bases then close the infinite towers and their ramification filtration. These classical source mechanisms are not a target Riemann construction. The audit records exact finite regressions separately from the all-parameter proof; no finite evidence or formal determinant upgrades the target claim.

References

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